Section 2

Types of microbial culture media

Types of culture media based on chemical composition/application:

There are seven routine laboratory media.

1. Basal media
2. Enriched media
3. Selective media
4. Enrichment media
5. Indicator media or differential media.
6. Transport media
7. Storage media.

Classification of microbial media according to their chemical composition

1. Chemically defined media:

It consists of mineral salts, and includes the general needs of microbes from carbon compounds, energy and other growth requirements.

2. Chemically undefined media

a- **Peptone water medium**, which is a liquid medium which growth appears in the form of turbidity in the medium.

b- **The nutrient agar,** which is a solid medium and contains agar for the enrichment of wide range of bacteria, some types of fungi and yeasts grow on it. The nutrient agar medium is the same composition as the previous environment, with the addition of 20 g of agar / liter.

Types of microbial media according to texture:

a) **Natural Solid medium:** such as potato chips or carrot environments. The solid medium creates a flat surface that helps bacteria grow in the form of individual colonies that are easy to isolate in a pure condition.

b) **Solid medium:** It is for the isolation of bacteria as a pure culture on a solid medium. Robert Koch realized the use of solid media. Agar is used to hardening the media at 1.5- 2.0% concentration. Solid media allows the growth of bacteria as colonies by streaking on the medium. It can be used at 37 degrees Celsius. Colonies identification is done on this medium. Examples of Solid Media: Nutrient agar, MacConkey agar, Blood agar, Chocolate agar. Growth of bacteria on solid medium appear as smooth, rough, mucoid, round, irregular, filamentous, punctiform.

c) **Semi-solid medium:** This media shows **the motility of bacteria** and the cultivation of microaerophilic bacteria. **This media has agar at a concentration of 0.5% or less**. It has a jelly consistency. Examples of Semi-solid media: Stuart's and Amies media, Hugh and Leifson's

oxidation fermentation medium, and Mannitol motility media. The growth of bacteria in semi-solid appears as a thick line in the medium.

d). **Liquid medium:** This media shows the growth of a large number of bacteria. It is called Broth that allows bacteria to grow uniformly with turbidity. The growth occurs at 37ºC in an incubator for 24hrs. Liquid media don't have the addition of agar; it is for fermentation studies. Examples of Liquid media Nutrient broth, Tryptic soy broth, MR-VP broth, phenol red carbohydrate broth. Growth of bacteria in liquid media- Turbidity is seen at the end of the broth.

Solidifying agents:

Some substances are added to the liquid culture medium to solidify that helps in the formation of individual colonies.

Gelatin: A proteinous substance prepared by treating animal bones.

Agar: A carbohydrate substance extracted from some red marine algae, and it solidifies at a temperature of 42-45 °C, and it can be melted again at a temperature of 98 °C.

Classification microbial media according to their purpose:

1**. Rich media:** They are simple media in addition to rich nutrients such as peptone, yeast extract and beef extract.

a. **Blood agar medium:** It is a nutritious agar medium that contains 10% sheep or human blood. This medium is suitable for pathogenic bacteria, such as the genus Streptococcus.

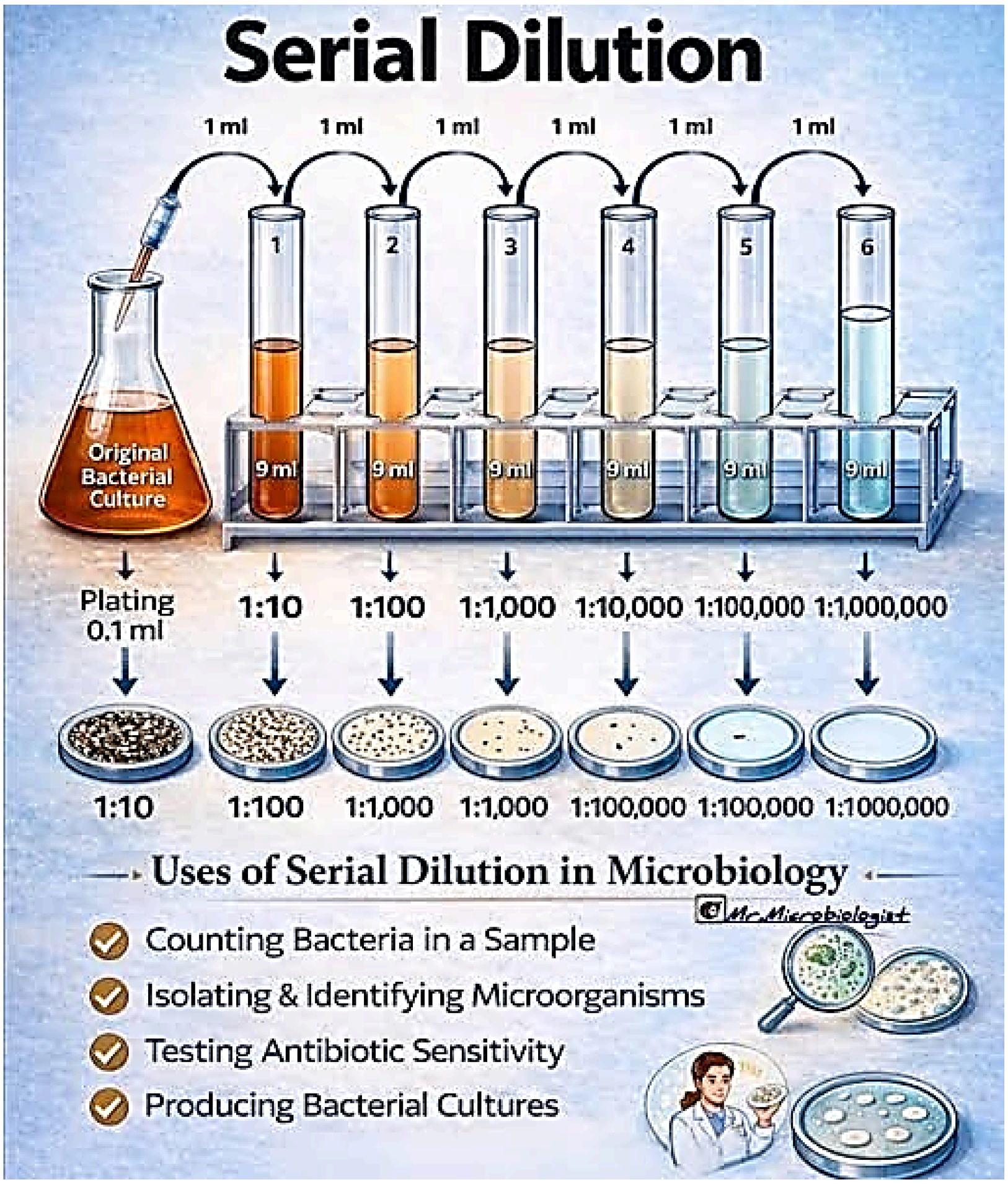
b - **Chocolate agar medium**, consist of blood agar medium, and chocolate. These components require types of bacteria so that they can grow, for example, Haemophilus influenza bacteria

2. **Selective media:** contains a substance that **inhibits** the growth of some types of bacteria while such as Dyes, bile salts, antibiotics, acids to allow the growth of a group of bacteria without others.

3. **Differential media:** They are the media that allow the growth of two types of bacteria that can be distinguished between them

a- **Blood agar environment** it allows distinguishing hemolytic and non-hemolytic bacteria, as an empty ring appears around the dissolved colony.

b- **Eosin methylene blue medium** is used to distinguish E.coli colonies with black centers and greenish metallic luster from other coliform bacteria.


Serial Dilution
1 ml
1 ml
1 ml
1 ml
1 ml
1 ml
1
2
3
4
5
6
Original Bacterial Culture
9 ml
9 ml
9 ml
9 ml
9 ml
9 ml
Plating 0.1 ml
1:10
1:100
1:1,000
1:10,000
1:100,000
1:1,000,000
1:10
1:100
1:1,000
1:1,000
1:100,000
1:100,000
1:1000,000
Uses of Serial Dilution in Microbiology
Counting Bacteria in a Sample
Isolating & Identifying Microorganisms
Testing Antibiotic Sensitivity
Producing Bacterial Cultures
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